Resistive switching meets photonics

Electrical switching can change the optical response of a waveguide or resonator. This unit separates the measured electrical state, optical transmission, the physical mechanism proposed for their coupling, and retention after the voltage is removed.

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What is a plasmonic mode?

A surface plasmon polariton (SPP) is an electromagnetic mode coupled to collective charge motion at a conductor–dielectric interface. Its field can be confined near the interface, allowing a compact active region to interact with light.

Confinement is not free: the field penetrating a lossy conductor dissipates power. The best device balances overlap, attenuation, mode conversion, bandwidth and fabrication constraints for a specified geometry and wavelength.

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A planar interface as a model

For two homogeneous, isotropic nonmagnetic half-spaces with complex relative permittivities ϵ_m and ϵ_d, the ideal planar-interface transverse-magnetic (TM) surface mode has

$$\beta = k_0 \sqrt{\frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}}, \qquad k_0 = \frac{2\pi}{\lambda_0}.$$

Choose a decaying field branch on each side. A practical metal has complex ϵ_m, so β is generally complex: tighter confinement and higher loss often compete. This formula is not the dispersion relation of every finite, multilayer or patterned structure.

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Deriving the TM boundary condition

Put the interface at $z = 0$, dielectric in $z > 0$ and metal in $z < 0$. For propagation along x with the $e^{-i\omega t}$ convention, use a TM field with $H_y \propto e^{i\beta x - \kappa_d z}$ above and $H_y \propto e^{i\beta x + \kappa_m z}$ below. The physical surface-mode branches satisfy $\operatorname{Re}\kappa_d > 0$ and $\operatorname{Re}\kappa_m > 0$.

In each homogeneous half-space Maxwell's equations give $\kappa_j^2 = \beta^2 - \epsilon_j k_0^2$. Continuity of H_y and tangential E_x, using $\nabla \times H = -i\omega\epsilon_0\epsilon_j E$, yields

$$\frac{\kappa_d}{\epsilon_d} + \frac{\kappa_m}{\epsilon_m} = 0.$$

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Solve, then check the branch

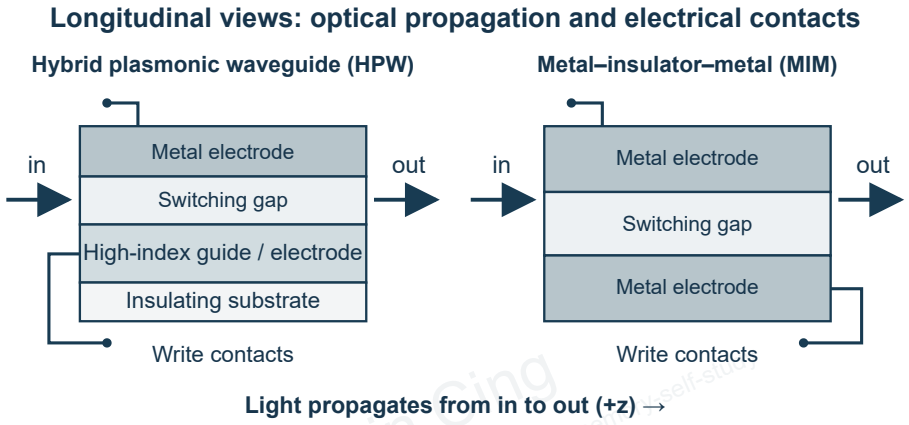
Substituting $\kappa_j^2 = \beta^2 - \epsilon_j k_0^2$ into the squared boundary equation gives

$$\beta^2 = k_0^2 \frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}.$$

Squaring can introduce an extraneous solution: substitute the selected square roots back into the **unsquared** boundary condition and verify decay on both sides. For the ideal lossless case with $\epsilon_d > 0$, a bound solution requires $\epsilon_m < -\epsilon_d$. A lossy, finite-thickness or MIM guide needs its own complex dispersion and physical branch selection.

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Two waveguide families



HPW example: conductive high-index guide (e.g., ITO); schematic.

A hybrid plasmonic waveguide (HPW) uses a narrow low-index gap between a high-index guide and metal; a metal–insulator–metal (MIM) structure supports gap-confined modes between conductors. Geometry and surrounding materials determine the mode; the sketches are conceptual longitudinal views, not fabrication recipes.

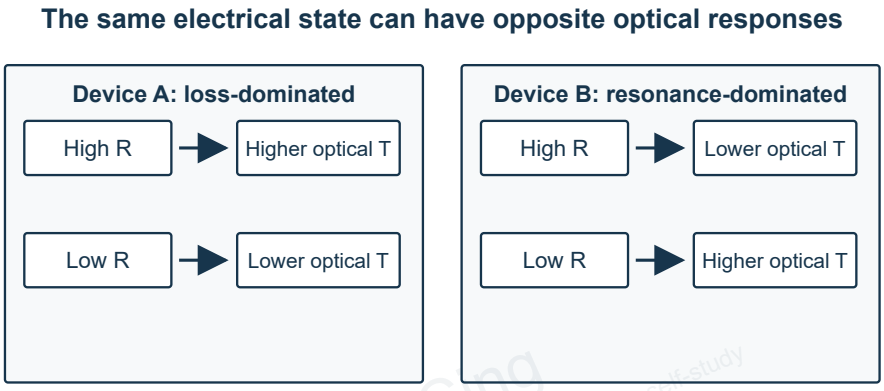
How a resistive state can alter light

The active region may change local complex permittivity, absorption, scattering, carrier density, boundary geometry or a resonance. These effects can coexist. Electrical and optical changes occurring together support a coupling hypothesis but do not alone identify the atoms or defects responsible.

Compare measurements at matched wavelength, input power, polarization and temperature, with a stated reference for normalization. Monitor both transmitted and reflected or scattered light when possible.

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Electrical labels do not determine optical labels



Optical transmission depends on the mode and material, not on resistance alone.

One structure may have lower transmission in its low-resistance state because of increased absorption; another can have higher transmission because a conductive bridge changes its resonance. LRS and HRS are electrical labels. Their optical ordering is a measured property of a particular device, wavelength and geometry.

From a complex effective index to power

Assume a single forward mode, fixed coupling, no reflection or conversion, and the time convention $E \propto e^{i\beta z - i\omega t}$ with $n_{\mathrm{eff}} = n + i\kappa$, $\kappa \geq 0$. Then $\beta = k_0(n + i\kappa)$ and the field magnitude falls as $e^{-k_0 \kappa L}$.

Since power is proportional to squared field magnitude under this normalization,

$$T = \frac{P(L)}{P(0)} = \exp\left(-\frac{4\pi\kappa L}{\lambda_0}\right).$$

Here κ belongs to the **mode**, not automatically to a bulk-material index. Reflections, mode mismatch, scattering and state-dependent coupling require a fuller model.

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A numerical two-state attenuation example

Let $\lambda_0 = 1.55\,\mu\mathrm{m}$ and $L = 2\,\mu\mathrm{m}$. For hypothetical $\kappa_H = 0.01$ and $\kappa_L = 0.08$, the exponents are -0.1621 and -1.2972; therefore $T_H \approx 0.8503$ and $T_L \approx 0.2733$.

These values are conditional on the single-mode assumptions. They do not describe an experimentally measured component or include coupling loss.

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Extinction ratio and insertion loss

With the same input power in both states, define $ER = 10\log_{10}(T_H/T_L)$ and high-transmission-state insertion loss $IL_H = -10\log_{10} T_H$; state the reference plane and any coupling losses.

For the preceding example, $ER \approx 4.929\,\mathrm{dB}$ and $IL_H \approx 0.704\,\mathrm{dB}$. Large extinction does not by itself guarantee low insertion loss or low switching energy.

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Cascading ideal stages

Three identical independent stages with unchanged coupling have $T_{\mathrm{total}} = T_H^3 \approx 0.615$ and $IL_{\mathrm{total}} = 3IL_H \approx 2.112\,\mathrm{dB}$. Losses add in dB because linear transmission factors multiply.

Reflections, coherent interference, amplifier noise and extra couplers invalidate this simple multiplication. Measure system loss at consistent input and output reference planes.

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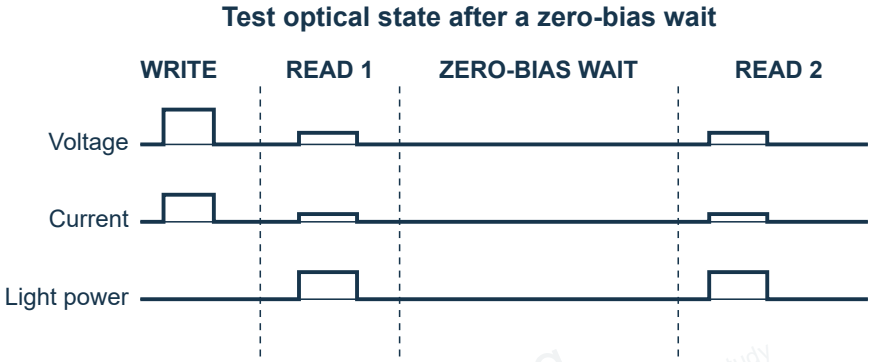
Percentages and decibels are different questions

Suppose a reflection changes from $R_1 = 0.20$ to $R_2 = 0.25$. The absolute change is five percentage points. The ratio is $R_2/R_1 = 1.25$, giving $10\log_{10}(1.25) \approx 0.969\,\mathrm{dB}$.

If only the five-point difference is known, the dB ratio is not determined: a different baseline produces a different ratio. Specify whether reflectance, transmission or a loss is reported.

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Separate writing, reading and waiting



Define the voltage at the device, current, optical probe, timing reference and read disturbance. A practical sequence is write → low-disturbance optical read → remove the bias → wait at controlled temperature → read again.

A hysteresis loop or optical contrast while voltage is applied does not establish zero-bias nonvolatile retention. Repeat after specified wait times and across devices and cycles.

Electrical energy is an integral

Electrical energy delivered to the device over a chosen interval is $E_{\mathrm{dev}} = \int_{t_0}^{t_1} V_{\mathrm{dev}}(t) I_{\mathrm{dev}}(t)\, dt$. Define current direction and measure at the device, since leads and parasitic capacitors can add loss or stored energy.

For an ideal rectangular pulse, $V = 1.2\,\mathrm{V}$, $I = 10\,\mu\mathrm{A}$ and $\Delta t = 20\,\mathrm{ns}$ give $E_{\mathrm{dev}} = 240\,\mathrm{fJ}$. Drivers, reset, sensing, lasers and interconnect can increase full-system energy.

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Power divided by rate needs assumptions

If the average power over a complete observation period, including active and idle intervals, is $P_{\mathrm{avg}} = 12.5\,\mathrm{nW}$ and the average rate is one operation per period at $f = 1\,\mathrm{MHz}$, then $P_{\mathrm{avg}}/f = 12.5\,\mathrm{fJ}$ per operation on average. Instantaneous power need not be constant; the duration, power measurement boundary and definition of an operation matter.

Do not substitute peak power, active-only power or an unrelated maximum frequency into P/f and call it a measured write energy. The average includes whatever idle energy lies within the stated boundary and need not equal one individual write pulse's energy. Integrate synchronized waveforms to distinguish them.

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Historic example: hybrid waveguide

A historical ITO/SiO_2/Au hybrid-waveguide example in the reference reports a roughly $10\ \mu\mathrm{m}$ active length, operation around $\pm 2\ \mathrm{V}$, optical contrast near $12\ \mathrm{dB}$ and a $30\ \mathrm{MHz}$ figure. These belong to a particular device and test configuration, not universal specifications.

The reported $200\ \mathrm{nW}$ is not automatically an energy per operation. Filament composition and VCM/ECM contributions require evidence beyond concurrent current and optical changes.

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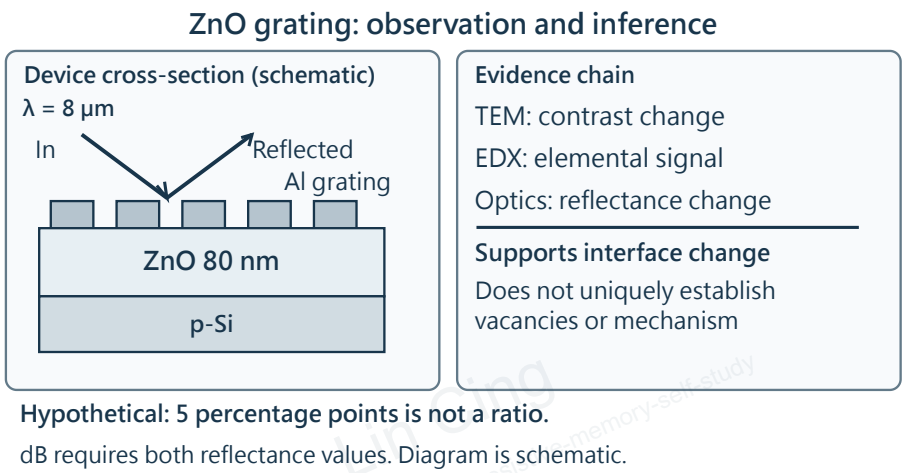
A thermal device can be volatile

The reference discusses a VO_2 waveguide in which heating changes the optical state. A thermal mechanism can return toward its original state after electrical power is removed; a sustained bias or other energy input may be necessary.

The reference's body and figure description differ in stated device lengths. Its figure description distinguishes about $1\ \mu\mathrm{m}$ with $12\ \mathrm{dB}$ and $0.5\ \mu\mathrm{m}$ with $4\ \mathrm{dB}$; these are separate devices, not a single combined performance specification.

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ZnO grating: the measurement configuration



Hypothetical: 5 percentage points is not a ratio.

dB requires both reflectance values. Diagram is schematic.

The reference describes a patterned Al electrode on an approximately $80\,\mathrm{nm}$ ZnO layer on heavily doped p-Si. The electrical bias is applied between the Al pad and Si substrate; the grating is illuminated at $8\,\mu\mathrm{m}$ and reflected light is recorded during voltage sweeps.

What the optical and chemical measurements establish

For that particular sample the review reports two electrically associated reflection states and a difference of about 5% in reflection. The observed low-resistance state corresponds to the higher-reflection state in that experiment. A reflection state was observed at zero bias; the report alone does not specify an indefinitely long retention time.

TEM contrast and an EDX line scan showing a Zn–O composition difference are consistent with local material changes. The reference describes Zn–O concentration differences of 5% in LRS and 1.85% in HRS for this particular measurement; these are not direct oxygen-vacancy concentrations. The authors interpret the observations as evidence for an oxygen-migration-related conductive path. Contrast is not a direct vacancy map: thickness, sample preparation, local chemistry and analysis uncertainty are relevant alternatives or confounders.

Reflection difference is not a dB ratio

An absolute reflectance difference in **percentage points** needs a baseline before it can be converted to a power ratio. For a hypothetical comparison $R_1 = 0.30$ and $R_2 = 0.35$, the difference is 0.05, or five percentage points, whereas $10 \log_{10}(0.35/0.30) \approx 0.669\,\mathrm{dB}$.

The reference reports approximately 5% and approximately $0.35\,\mathrm{dB}$ for its device; the precise percentage convention and both raw levels must be checked before reproducing that conversion. The numbers in the calculation above are **hypothetical**, not measurements of the ZnO device. Keep the wavelength, polarization, illumination and reflectance reference fixed.

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A historic a-Si waveguide: optical decrease

An Ag/amorphous-Si hybrid-waveguide example reports a low-resistance state associated with greater absorption and lower optical transmission during a slow sweep from -3 to $+9\,\mathrm{V}$ with approximately $1.5\,\mu\mathrm{m}$ light; the reported modulation is approximately 1%. The electrical and optical traces are observations under that sweep.

The observation alone does not show how long the optical state persists after voltage removal. A matched write–read–zero-bias-wait protocol is needed to characterize retention.

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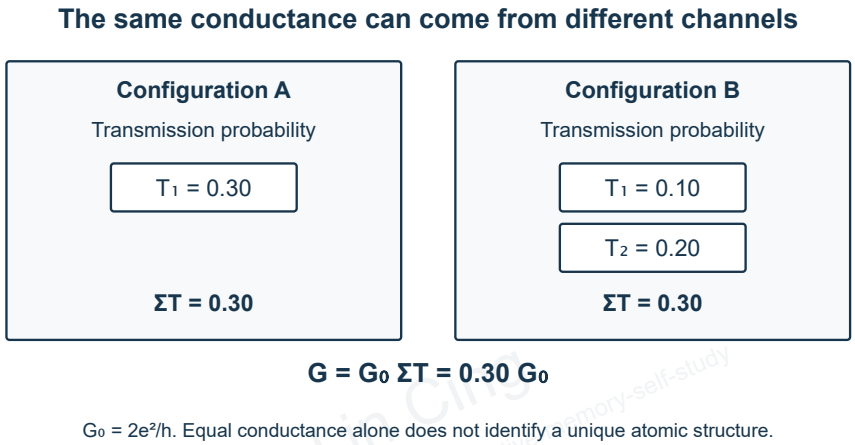
An atomic-scale switch: opposite optical ordering

Another Ag/a-Si MIM configuration reports increased optical transmission in an electrically conducting state. The reference describes an approximately $20\,\mathrm{nm}$ gap and a $1\,\mathrm{kHz}$ sweep with SET near $+1.25\,\mathrm{V}$ and RESET near $+0.25\,\mathrm{V}$ under its bias convention. Its mode and resonant response differ from the previous waveguide, illustrating why an electrical label cannot predict optical sign.

The original study reports $9.2\,\mathrm{dB}$ contrast under its stated conditions (Nano Letters, DOI 10.1021/acs.nanolett.5b04537). The reference gives differing contrast figures in its text and table; use the primary paper's figure with its conditions rather than merging them into an averaged specification. A sweep or biased hold does not by itself establish a particular zero-bias retention time.

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What a conductance step can—and cannot—show



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At low temperature and small bias in an ideal coherent transport model,

$$G = G_0 \sum_n T_n, \qquad G_0 = \frac{2e^2}{h} \approx 77.5\,\mu\mathrm{S},$$

Here $0 \le T_n \le 1$ are channel transmission probabilities.

A conductance calculation and its limits

For one hypothetical channel with $T_1 = 0.3$, $G = 0.3G_0 = 23.25\,\mu\mathrm{S}$ and $R = 1/G \approx 43.0\,\mathrm{k\Omega}$. Two channels with $T_1 = 0.1, T_2 = 0.2$ have the same total in this ideal model. A step can support a small-channel interpretation, but is not a unique atomic photograph.

Finite temperature, contact resistance, inelastic processes, bias dependence and optical field redistribution complicate inference from a measured conductance plateau.

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Quantify a no-failure observation honestly

Suppose 100 independent trials under the **same stated retention time and conditions** show zero failures. For a common failure probability p, $P(0\ \mathrm{failures}) = (1-p)^{100}$.

Setting this probability to 0.05 gives the one-sided 95% upper bound $p_{\mathrm{upper}} = 1 - 0.05^{1/100} \approx 0.0295$, or 2.95%. The bound relies on independent, identically distributed trials: correlated cycles or devices violate the calculation. It says nothing about untested wait times or conditions.

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How to compare reports

For every number, record the structure, state definition, wavelength, polarization, bias, pulse shape, temperature, bandwidth, number of samples and optical reference. Distinguish a stated pulse-integrated energy from a power, and distinguish zero-bias wait from a biased hold.

Optical contrast, switching speed, endurance and retention measured on different devices or in different test runs cannot automatically be combined into one guaranteed operating point.

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Design a discriminating experiment

Use matched electrical and optical time traces with known parasitics. Compare polarization and wavelength dependence, contact materials and controls; probe temperature and zero-bias retention; combine optical data with spatial or chemical measurements when feasible.

If several mechanisms fit the same traces, report the competing explanations and what observation could distinguish them. A correlation or a fitted curve alone rarely determines a unique microscopic mechanism.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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